

Efficiency Corrections Comparison

Current vs. Previous Results

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Methodology

Reconstruction Efficiency and Covariance

1. Mean Reconstruction Efficiency: Calculated as the arithmetic mean of event-by-event interpolated efficiencies for N events in a bin:

$$\bar{E}_{\text{reco}} = \frac{1}{N} \sum_{i=1}^N E_{\text{reco},i}$$

2. Correlated Error Propagation: Because efficiencies are interpolated from the same D_1 nodes, event uncertainties (σ_i) are correlated. We construct a covariance matrix:

$$Cov_{i,j} = \rho_{i,j} \sigma_i \sigma_j$$

- $\rho = 1$ if events share identical or adjacent D_1 occupancy.
- $\rho = 0$ if events use entirely different nodes.

$$\sigma_{\bar{E}_{\text{reco}}} = \frac{1}{N} \sqrt{\sum_{i=1}^N \sum_{j=1}^N Cov_{i,j}}$$

- 1. Hodoscope Mean Efficiency:** Evaluated with purely independent, standard weighted uncertainty:

$$\sigma_{\bar{E}_{\text{hodo}}} = \frac{1}{N} \sqrt{\sum_{i=1}^N \sigma_{\text{hodo},i}^2}$$

- 2. Final Bin Efficiency:** Computed as the product of the independent bin averages:

$$\bar{E}_{\text{final}} = \bar{E}_{\text{reco}} \times \bar{E}_{\text{hodo}}$$

- 3. Final Propagated Uncertainty:**

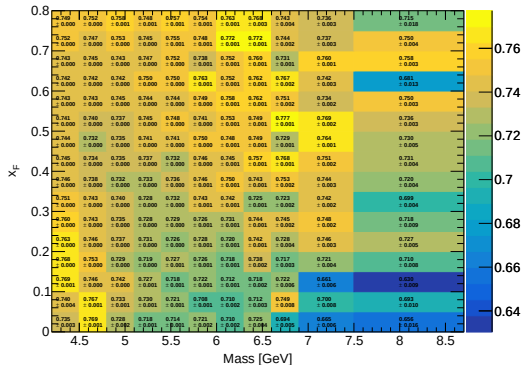
$$\sigma_{\bar{E}_{\text{final}}} = \sqrt{(\bar{E}_{\text{hodo}} \cdot \sigma_{\bar{E}_{\text{reco}}})^2 + (\bar{E}_{\text{reco}} \cdot \sigma_{\bar{E}_{\text{hodo}}})^2}$$

LH2 Efficiency Corrections

LH2 Target: Total Yield

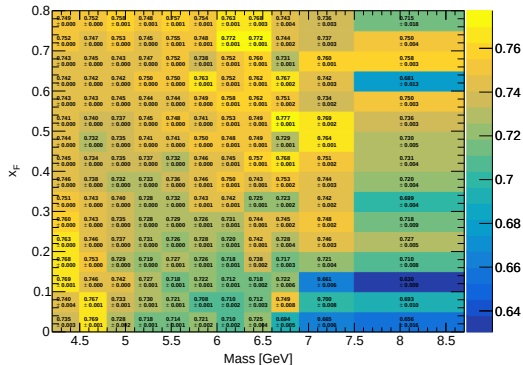
Hodoscope Efficiency Corrections

Avg Hodo Eff (Total) (LH2)



(a) Previous

Avg Hodo Eff (Total) (LH2)

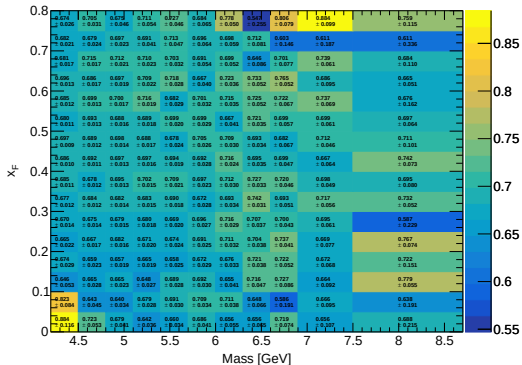


(b) Current

LH2 Target: Total Yield

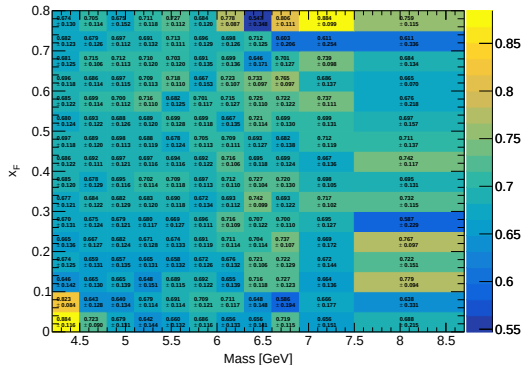
Reconstruction Efficiency Corrections

Avg Reco Eff (Total) (LH2)



(a) Previous

Avg Reco Eff (Total) (LH2)

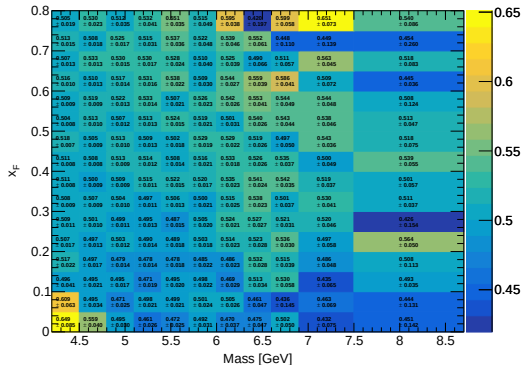


(b) Current

LH2 Target: Total Yield

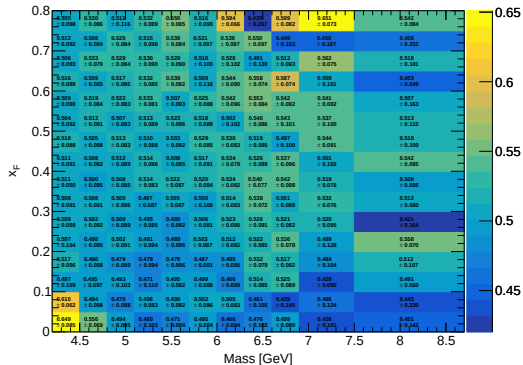
Final Efficiency Corrections

Avg Final Eff (Total) (LH2)



(a) Previous

Avg Final Eff (Total) (LH2)

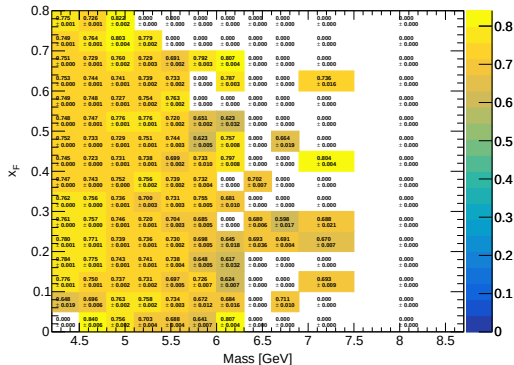


(b) Current

LH2 Target: Mix Yield

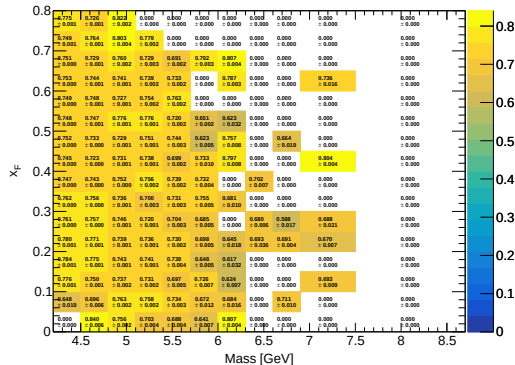
Hodoscope Efficiency Corrections

Avg Hodo Eff (Mix) (LH2)



(a) Previous

Avg Hodo Eff (Mix) (LH2)

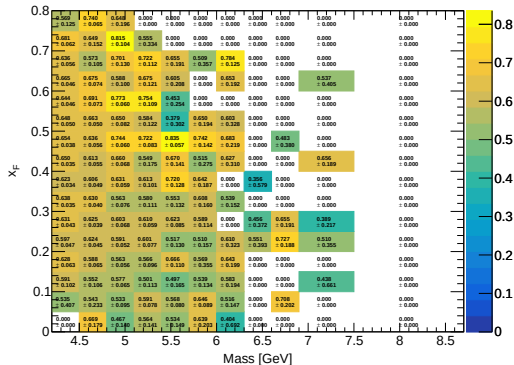


(b) Current

LH2 Target: Mix Yield

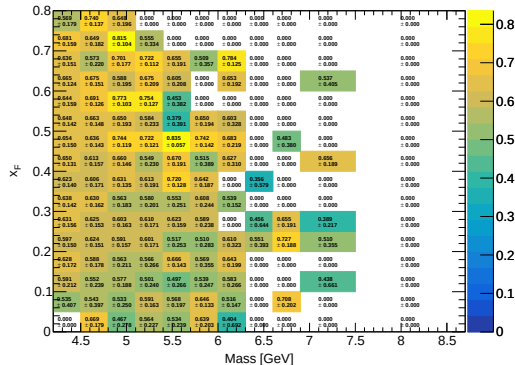
Reconstruction Efficiency Corrections

Avg Reco Eff (Mix) (LH2)



(a) Previous

Avg Reco Eff (Mix) (LH2)

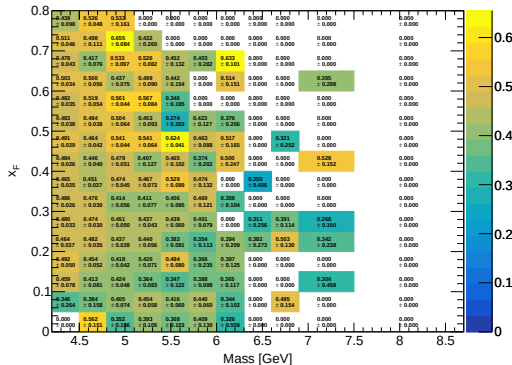


(b) Current

LH2 Target: Mix Yield

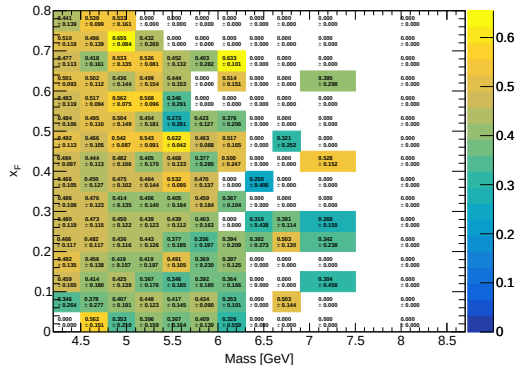
Final Efficiency Corrections

Avg Final Eff (Mix) (LH2)



(a) Previous

Avg Final Eff (Mix) (LH2)

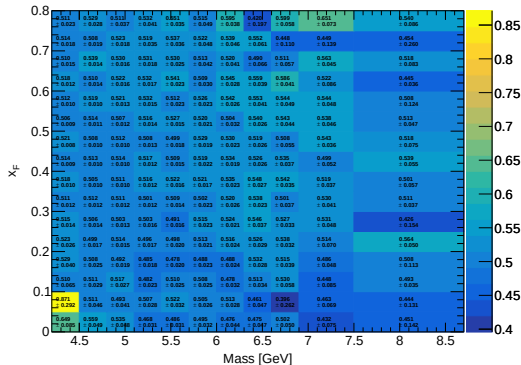


(b) Current

LH2 Target: Signal Efficiency

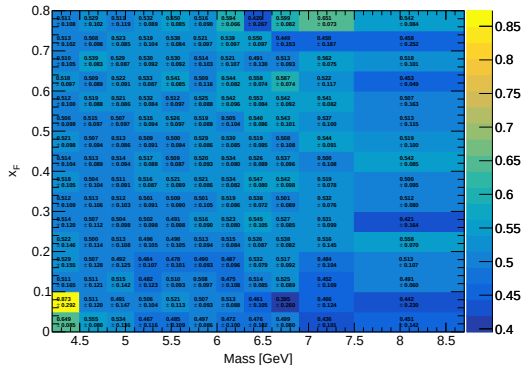
Final Corrections

Avg Signal Efficiency (LH2)



(a) Previous

Avg Signal Efficiency (LH2)



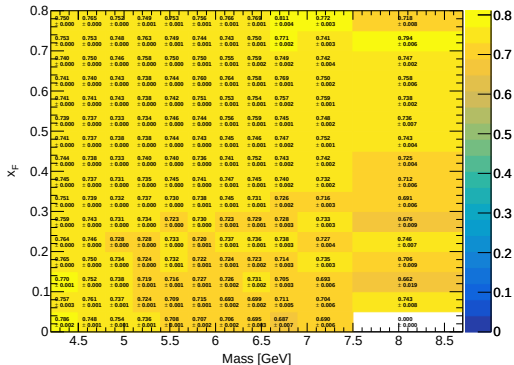
(b) Current

LD2 Efficiency Corrections

LD2 Target: Total Yield

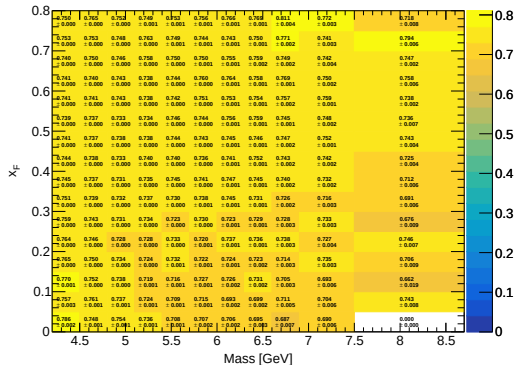
Hodoscope Efficiency Corrections

Avg Hodo Eff (Total) (LD2)



(a) Previous

Avg Hodo Eff (Total) (LD2)

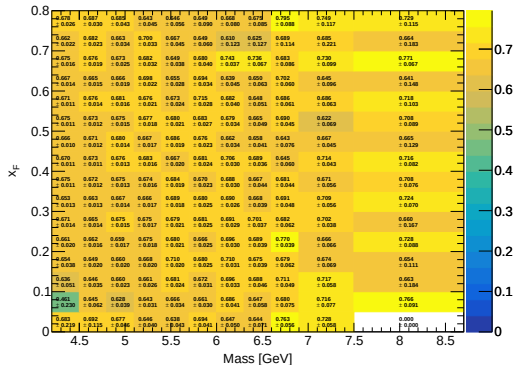


(b) Current

LD2 Target: Total Yield

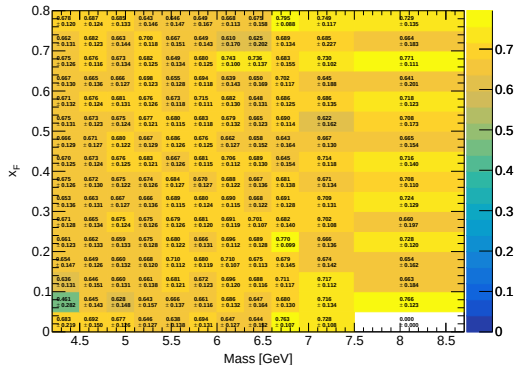
Reconstruction Efficiency Corrections

Avg Reco Eff (Total) (LD2)



(a) Previous

Avg Reco Eff (Total) (LD2)

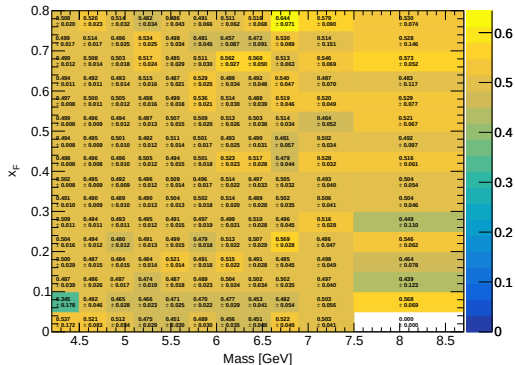


(b) Current

LD2 Target: Total Yield

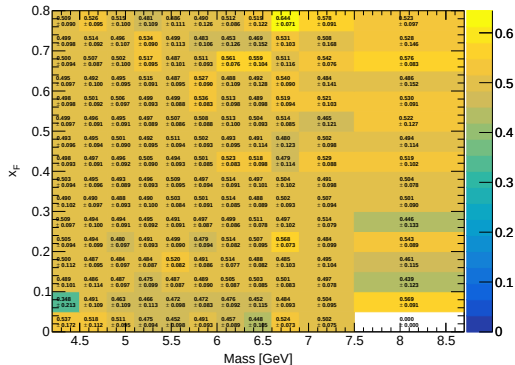
Final Efficiency Corrections

Avg Final Eff (Total) (LD2)



(a) Previous

Avg Final Eff (Total) (LD2)

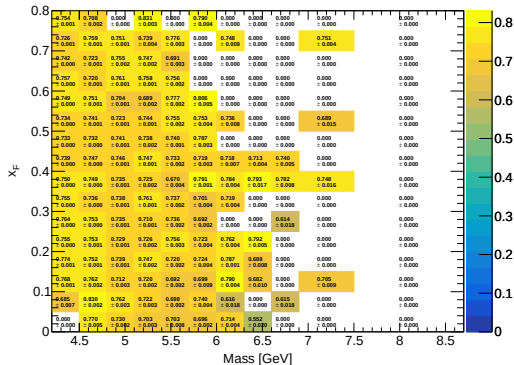


(b) Current

LD2 Target: Mix Yield

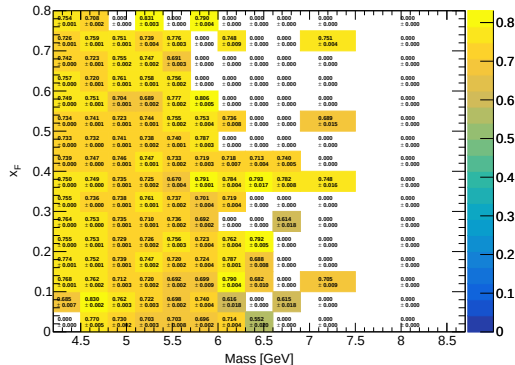
Hodoscope Efficiency Corrections

Avg Hodo Eff (Mix) (LD2)



(a) Previous

Avg Hodo Eff (Mix) (LD2)

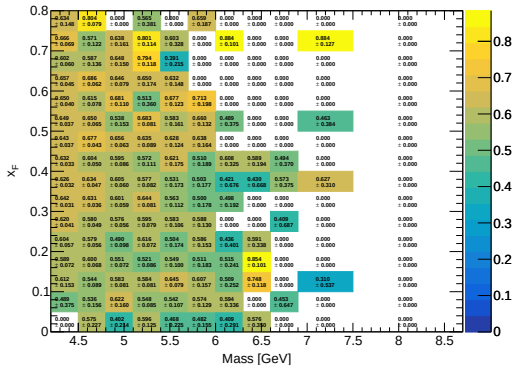


(b) Current

LD2 Target: Mix Yield

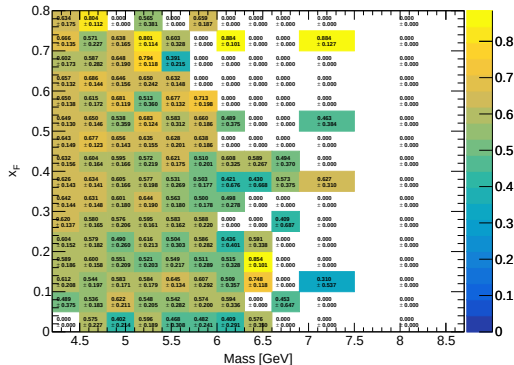
Reconstruction Efficiency Corrections

Avg Reco Eff (Mix) (LD2)



(a) Previous

Avg Reco Eff (Mix) (LD2)

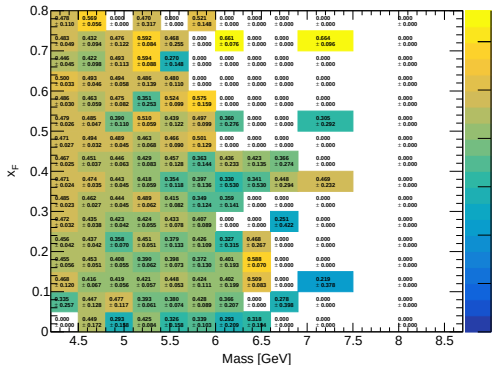


(b) Current

LD2 Target: Mix Yield

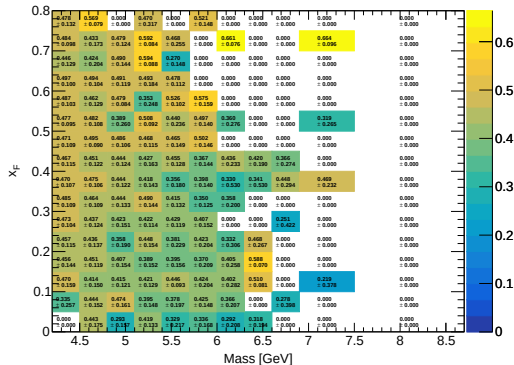
Final Efficiency Corrections

Avg Final Eff (Mix) (LD2)



(a) Previous

Avg Final Eff (Mix) (LD2)

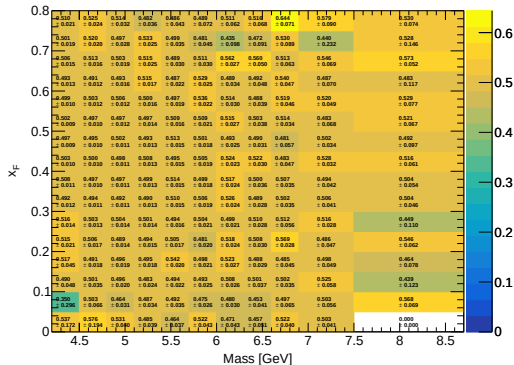


(b) Current

LD2 Target: Signal Efficiency

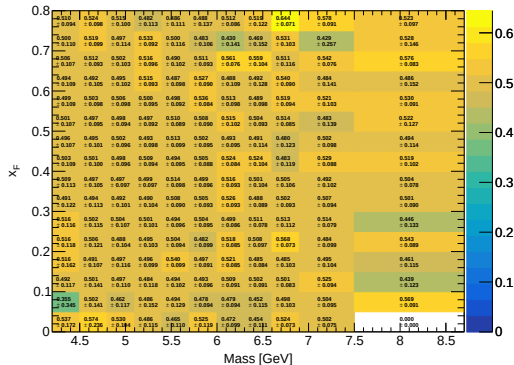
Final Corrections

Avg Signal Efficiency (LD2)



(a) Previous

Avg Signal Efficiency (LD2)



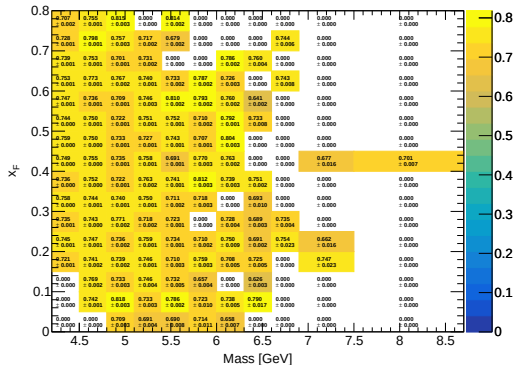
(b) Current

Flask Efficiency Corrections

Flask Target: Total Yield

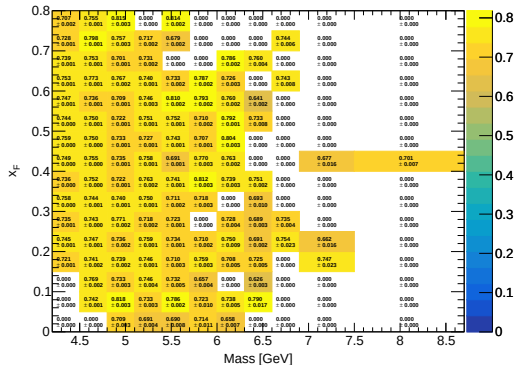
Hodoscope Efficiency Corrections

Avg Hodo Eff (Total) (Flask)



(a) Previous

Avg Hodo Eff (Total) (Flask)

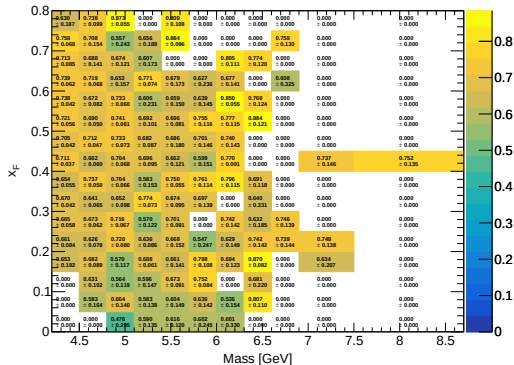


(b) Current

Flask Target: Total Yield

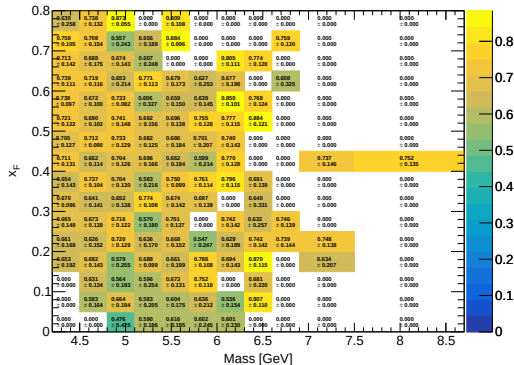
Reconstruction Efficiency Corrections

Avg Reco Eff (Total) (Flask)



(a) Previous

Avg Reco Eff (Total) (Flask)

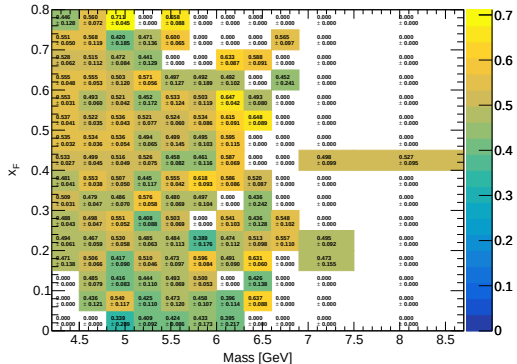


(b) Current

Flask Target: Total Yield

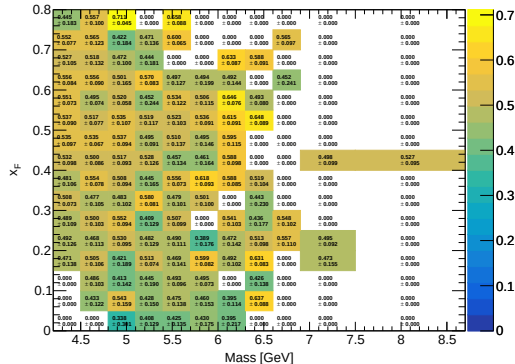
Final Efficiency Corrections

Avg Final Eff (Total) (Flask)



(a) Previous

Avg Final Eff (Total) (Flask)

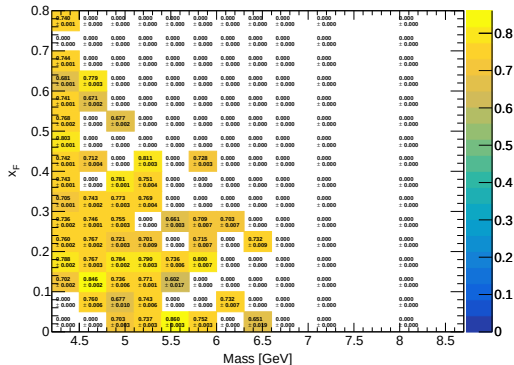


(b) Current

Flask Target: Mix Yield

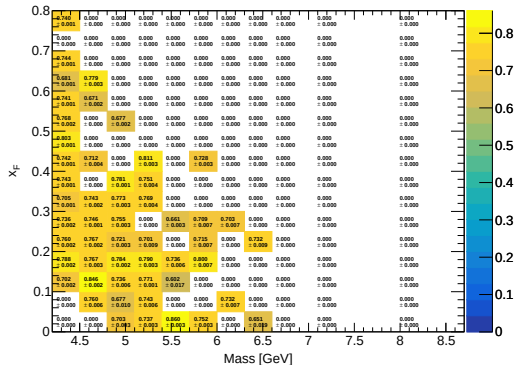
Hodoscope Efficiency Corrections

Avg Hodo Eff (Mix) (Flask)



(a) Previous

Avg Hodo Eff (Mix) (Flask)

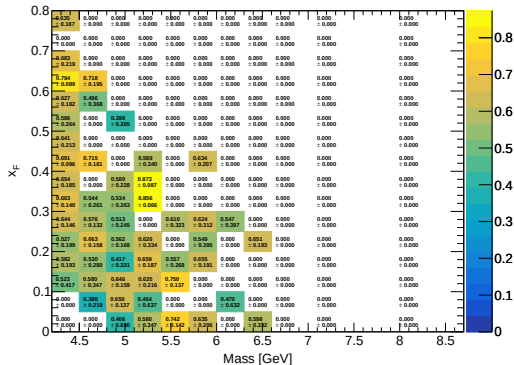


(b) Current

Flask Target: Mix Yield

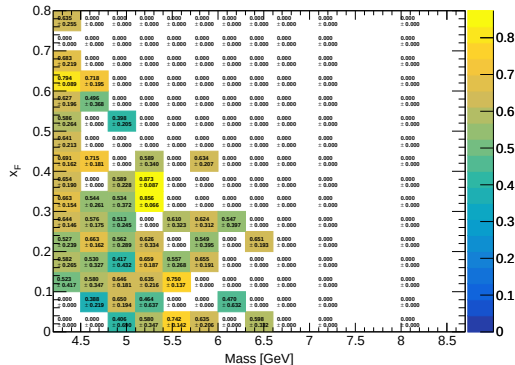
Reconstruction Efficiency Corrections

Avg Reco Eff (Mix) (Flask)



(a) Previous

Avg Reco Eff (Mix) (Flask)

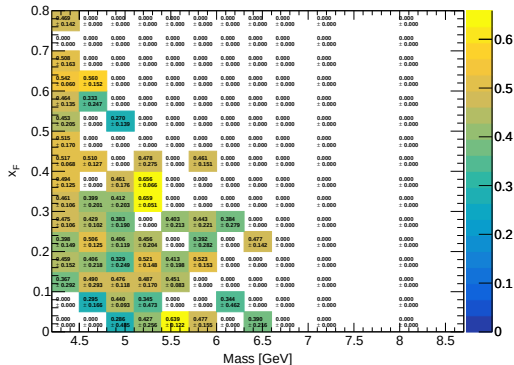


(b) Current

Flask Target: Mix Yield

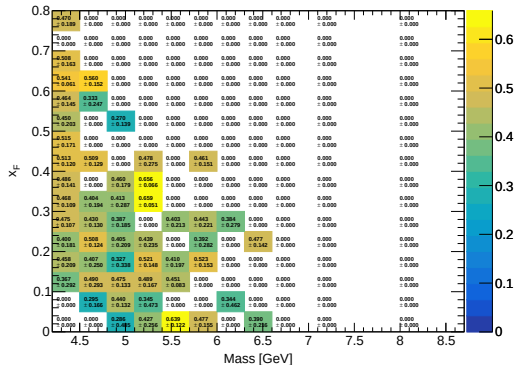
Final Efficiency Corrections

Avg Final Eff (Mix) (Flask)



(a) Previous

Avg Final Eff (Mix) (Flask)

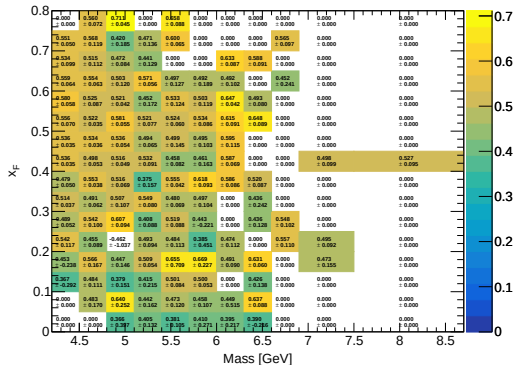


(b) Current

Flask Target: Signal Efficiency

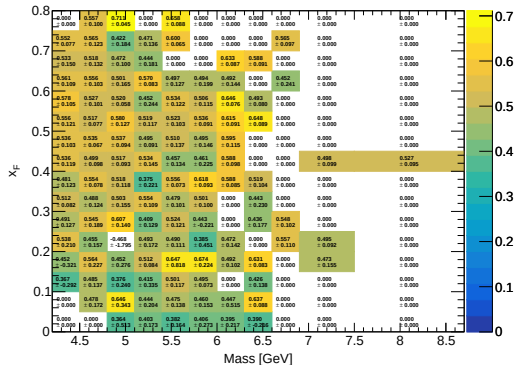
Final Corrections

Avg Signal Efficiency (Flask)



(a) Previous

Avg Signal Efficiency (Flask)



(b) Current